Fetosense OSS

Hardware Description — Reference Architecture, Protocol & Build Guide

Document	Fetosense OSS Hardware Description
Version	2.0 — Public Release Draft
Owner	CareNX Innovations
Audience	Embedded developers, biomedical engineers, researchers
Classification	Public (OSS) — Subject to third-party redistribution review (§0.3)
Repositories	Fetosense OSS Web/MIS · Remote · Device · Auto

0. Document Control

0.1 Revision History

Version	Date	Change	Author
1.0	—	Initial hardware outline	CareNX
2.0	—	Full rewrite: architecture, protocol, BOM, safety, images	CareNX

0.2 Purpose and Scope

This document enables a developer to understand the Fetosense OSS reference hardware sufficiently to **integrate with it, extend it, or build a functionally equivalent device** for research and non-clinical development. It describes:

- System topology and component roles
- Measured and specified performance of each transducer
- The full application-layer packet protocol between probes, receiving board, and host
- Power, battery, and environmental characteristics
- Operating workflow, charging, and maintenance
- Reference bill of materials
- Regulatory and safety context

0.3 Legal and Redistribution Notice

> **Read before redistributing.**
>
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1. System Overview and Architecture

1.1 Functional Summary

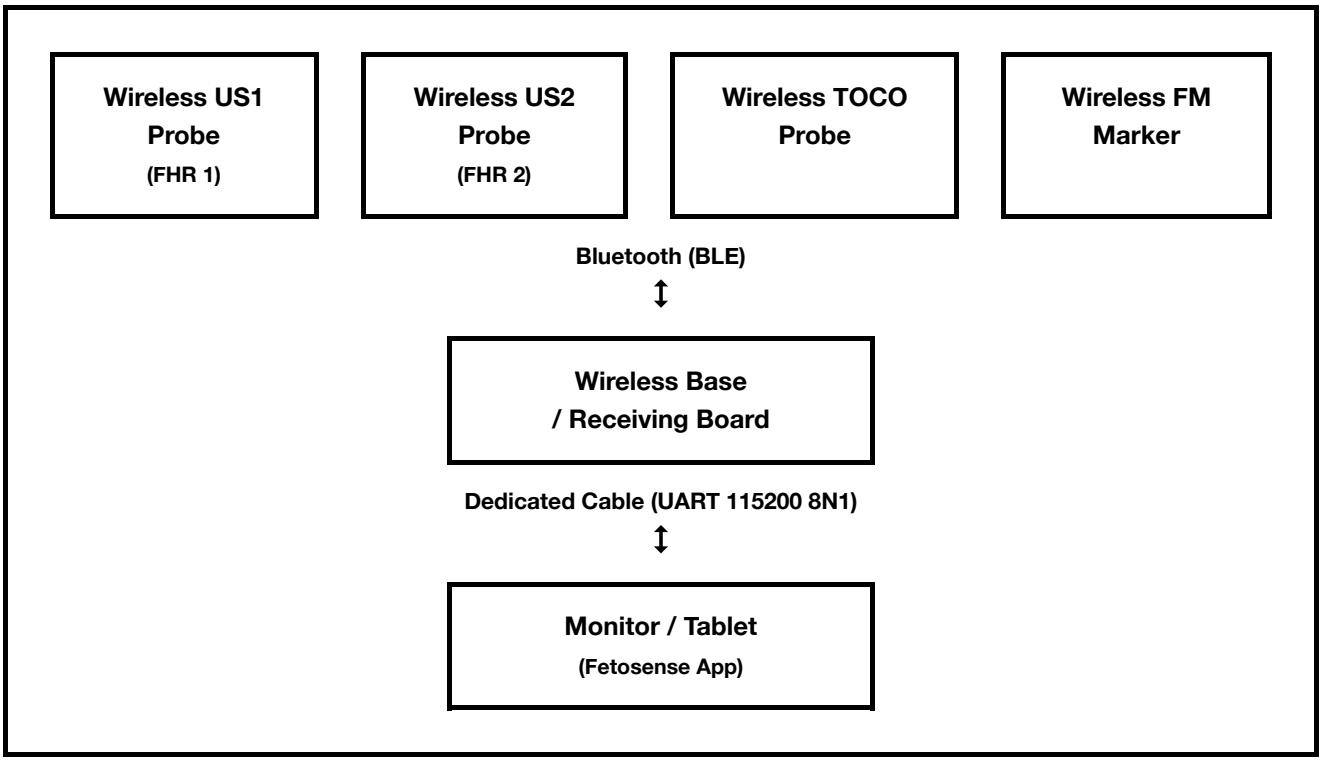
The Fetosense OSS reference kit is a **wireless, tablet-based cardiotocography (CTG / NST) system** that measures:

- **Fetal heart rate (FHR)** via pulsed Doppler ultrasound
- **Uterine contraction (TOCO)** via a tocodynamometer
- **Fetal movement (FM)** via a patient-operated event marker

The kit is portable, battery-operated, and designed for low-resource and point-of-care settings. It performs non-stress testing (NST) from the 28th week of gestation and is intended for use by trained personnel in antepartum examination rooms, gynecology clinics, pre-delivery rooms, and labor/delivery rooms.

1.2 Topology

The reference hardware uses a **receiving-board ("wireless base") architecture**. Probes communicate wirelessly to the base; the base connects to the monitor or tablet over a dedicated serial link.



Link summary:

Link	Physical layer	Settings / Notes
Host (tablet) ↔ Receiving board	UART over dedicated cable	115200 baud, 8 data bits, 1 stop bit, no parity
Receiving board ↔ Probe	Bluetooth (application packets defined in §3)	2.4 GHz ISM band, 0 dBm transmit power
Probe ↔ Receiving board (wired variant)	UART	19200 baud, 8N1

1.3 Wireless Characteristics

Parameter	Value
Band	2.4 GHz industrial / medical / scientific (ISM)
Transmit power	0 dBm
Working channels	101 (0–100)
Recommended channels	84–99 (to avoid WLAN interference)
Effective range	5 m
Room capacity	Up to 4 single-fetus terminals, or 1 twin + 2 single
Coexistence	Limit terminals per room; keep line-of-sight between probe and base

1.4 Kit Composition

Component	Function
Tablet	Runs the Fetosense Device application; displays CTG traces; stores and transmits data
Wireless US probe (FHR1)	Pulsed Doppler fetal heart rate measurement
Wireless US probe (FHR2, optional)	Second FHR channel for twin monitoring
Wireless TOCO probe	Uterine contraction measurement
Wireless FM marker	Patient-operated fetal movement event input
Wireless base / receiving board	Aggregates probe data; bridges to the host over serial; charges docked probes
Communication cable	Base ↔ monitor/tablet
Speaker	Audio output for fetal heart sound
Aluminum case	Portable kit enclosure

2. Component Descriptions and Specifications

2.1 Fetal Heart Rate (Ultrasound) Probe & TOCO Probe



FHR Probe (Right) + TOCO Probe (Left)

2.1.1 Measurement Principle

The US probe uses **pulsed Doppler ultrasound with autocorrelation processing**. An ultrasound beam (1.0 MHz or 2.0 MHz) is transmitted into the maternal abdomen. Reflections from moving structures — primarily fetal heart valves — are frequency-shifted per the Doppler effect. The probe demodulates the echoes to obtain differential frequencies representing motion within the beam, and the host computes the beat-to-beat interval to derive FHR.

2.1.2 Specification Table

Parameter	Value	Source
Transducer type	Multi-chip, wide-beam, pulsed Doppler	Device documentation
Operating frequency	1.0 MHz and/or 2.0 MHz (variant dependent)	Device documentation
FHR measurement range	50–210 bpm	Device documentation
FHR accuracy	±2 bpm (error); ±1 bpm resolution	Device documentation
Effective radiating area	549.5 mm² ±15% (7 crystals) to 1017 mm² ±15% (9 crystals)	Device documentation
Ultrasonic output intensity (Ispta)	≤ 5 mW/cm²	Device documentation
Acoustic output (IEC 1157:1992)	Negative peak ≤ 1 MPa; beam intensity ≤ 20 mW/cm²; spatial-peak temporal-average ≤ 100 mW/cm²	Device documentation
Probe type code (protocol)	`0x03` = FHR1, `0x08` = FHR2	§3.3
Display	0.96" color OLED, 128×64	Device documentation
Battery life	~11 h continuous monitoring	Device documentation
Charge time	~2 h	Device documentation
Couplant	Water-based only. Oil-based couplants damage the probe face and are prohibited.	Device documentation

2.2 Uterine Contraction (TOCO) Probe

2.2.1 Measurement Principle

A tocodynamometer measures abdominal wall tension changes associated with uterine contractions. The reference probe reports values in **pressure units (0–100)** rather than mmHg, and provides a **reset** function to establish a new baseline.

2.2.2 Specification Table

Parameter	Value	Source
Measurement range	0–100 pressure units	Device documentation
Non-linear accuracy	±10%	Device documentation
Reset values	0, 5, 10, 15, 20	§3.3, §3.5
Probe type code (protocol)	`0x0C`	§3.3
Battery life	~11 h continuous	Device documentation
Charge time	~2 h	Device documentation
Water resistance	**Non-waterproof.** Do not apply couplant; avoid liquid ingress.	Device documentation

2.3 Fetal Movement (FM) Event Marker



Movement Marker



Wireless Speaker

Parameter	Value
Form factor	Compact wireless handheld marker
Function	Patient presses on perceived fetal movement
Protocol signal	Sets Status BIT4 (manual MARKER) in monitoring packet
Pairing	Via base "FM setting" workflow (§6.3)
Auto-FM	The device also detects automatic fetal movement (Status BIT3), reported alongside the manual marker

2.4 Wireless Base (Receiving Board)

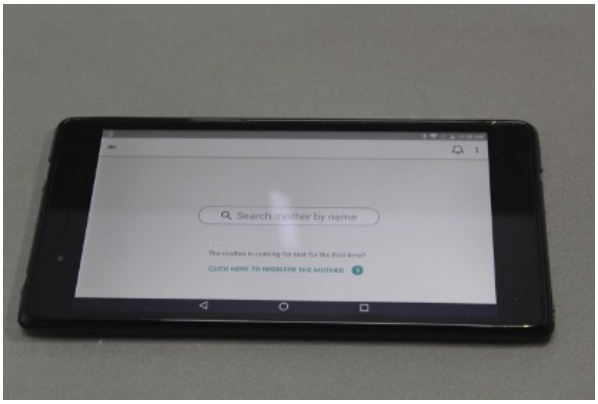
2.4.1 Role

The base is the **central aggregator**: it receives probe packets over BLE, timestamps and sequences them, and forwards them to the host over UART. It also **charges docked probes** and provides a local status UI.

2.4.2 Specification Table

Parameter	Value
Display	0.96" OLED, 128×64
Controls	On/Off, Setup, Up, Down, Back
Battery life (base)	~4.5 h
Charge time (base)	~4 h
Dimensions	210 (L) × 140 (W) × 115 (H) mm
Net weight	0.7 kg (including battery)
Ingress protection	IPX1 (drip-proof)
Charging indicators	Per-socket LED; lit while charging, off when complete

2.5 Monitor / Tablet & Charging



Tablet — Lenovo M8



Charging Cable and Adapter

Parameter	Value
Role	Runs Fetosense Device application
Connection to base	Dedicated communication cable (UART 115200 8N1)
Probe type selection	"Wireless" mode selected in the Fetal/Pregnant Woman menu
Display	Real-time FHR and TOCO traces, event markers, signal quality

3. Probe Communication Protocol

> **Attribution note.** This section summarizes an application-layer protocol document supplied by the reference hardware OEM. The protocol document itself is third-party proprietary material. Confirm redistribution rights before reproducing it in full.

3.1 Layering

Layer	Host ↔ Base	Base ↔ Probe
Physical	UART	BLE (or UART 19200 for wired variant)
Framing	Byte-oriented control words (§3.2)	Byte-oriented control words (§3.2)
Application	Monitoring, control, heartbeat	Monitoring, sound, control, pairing

3.2 Control Word Summary

3.2.1 Host → Base and Base → Host

ID	Meaning	Direction	Ack required
`0x01`	Set sound channel (0 mute, 1 FHR1, 2 FHR2)	Host → Base	Yes
`0x02`	Set sound volume (0–7)	Host → Base	Yes
`0x03`	TOCO reset (value 0/5/10/15/20)	Host → Base	Yes
`0x04`	Monitoring data	Base → Host	No
`0x05`	Packet count (per probe type)	Base → Host	No
`0x66`	Start / stop (0x00 start, 0xFF stop)	Host → Base	Yes
`0xA6`	Enter test mode	Host → Base	Yes
`0xA1`	Heartbeat / dock status	Base → Host	No
`0xC1`	Get info (index, bed, version, build time, MAC)	Host → Base	Yes
`0xC2`	Set parameters (pairing)	Host → Base	Yes
`0x08`	Pairing info (bed 0x16, ch1 0x17, ch2 0x18)	Host → Base	Yes
`0xD0`	Alarm status (level, volume)	Host → Base	No
`0xAA`	Heartbeat packet	Base → Host	No

3.2.2 Base → Probe and Probe → Base

ID	Meaning	Direction	Ack required
`0x04`	Monitoring packet	Probe → Base	Yes
`0x09`	Sound packet (ADPCM)	Probe → Base	No
`0x66`	Start / stop	Base → Probe	Yes
`0xC3`	Set options (bitfield, see §3.5)	Base → Probe	Yes
`0xB1`	Option return packet	Probe → Base	No
`0x52`	Monitoring data return	Base → Probe	No
`0x21`	First connection (probe power-up)	Probe → Base	Yes

Retry rule: Any packet requiring acknowledgement that receives no response within the timeout must be retransmitted.

3.3 Monitoring Packet (`0x04` / `0x52`)

Cadence: one packet per second per probe. On packet loss, the probe retransmits the lost data **every 100 ms**.

Byte offset	Field	Size	Description
0	Probe type	1	`0x03` FHR1, `0x08` FHR2, `0x0C` TOCO
1–2	Packet number	2	Little-endian (low byte first)
3–6	Data1	4	FHR in bpm (50–210; 0 when no FHR) **or** TOCO value
7–10	Data2	4	Automatic fetal movement curve data (FHR probes)
11	Status	1	Bitfield (see below)
12	Battery	1	0–100 (%)
13	Default	1	Fill 0

Status bitfield

Bit	Meaning
1–0	Signal quality: `01` poor, `10` fair, `11` good
2	TOCO reset (1 = reset)
3	Automatic fetal movement detected (1 = one movement)
4	Manual fetal movement marker pressed (1 = pressed)
5	External wake-up device activated (1 = active)
6	TOCO reset (duplicate indicator)
7	Reserved (0)

Packet numbering rules

- **Before start command:** packet number is always `0xFFFF`.
- **After start:** counter increments from 0; wraps to 0 after `0xFFFF`.
- **New probe joining mid-session:** its counter starts from the **highest current counter** among active probes.

3.4 Sound Packet (`0x09`)

Parameter	Value
Cadence	Every 50 ms
Retransmission	**None**
Encoding	ADPCM
Payload	Packet number (2 bytes) + 100 encoded sample bytes + checksum + ADPCM index/value
Decode	100 bytes → 200 PCM samples; decode per separate ADPCM reference document (see §11)

3.5 Control Packet Details

Start / Stop (`0x66`)

- 1 byte: `0x00` = start, `0xFF` = stop.
- Every start instruction **clears the packet counter**.

Set Options (`0xC3`, base → probe)

Bit	Meaning
0	Start monitoring
1	Stop monitoring
2	Send version number
3	Enter test mode
4	Turn on sound (US probes only)
5	Turn off sound (US probes only)
6	TOCO reset command (TOCO probe only)
7	Reserved (`0x02` — parameter byte follows)

A bit is **valid when set to 1**. The probe must return an option-return packet (`0xB1`); if not received, the base retransmits.

Option Return (`0xB1`, probe → base)

Bit	Meaning
0	Return of start/stop
1	Return of demo command
2	Return of US sound switch
3	Return of TOCO reset
4	Return of US volume setting
5–7	Reserved (0)

First Connection (`0x21`, probe → base on power-up)

No payload. The base replies with the current option block for that probe type:

US probe reply:

Byte	Bits	Meaning
1	BIT0	1 = test mode, 0 = normal
1	BIT1	1 = monitoring started
2	BIT0	Fetal heart sound channel
2	BIT7–4	Volume

TOCO probe reply:

Byte	Bits	Meaning
1	BIT0	1 = test mode
1	BIT1	1 = monitoring mode
2	—	Contraction reset value

Heartbeat (`0xA1`)

- Sent every 1 s by the probe (or by the base in non-monitoring mode).
- Includes probe status (docked, communicating), and battery levels for FHR1, FHR2, TOCO.
- The base ignores heartbeats whose bed number does not match the main Bluetooth terminal's bed.

Get Info (`0xC1`, host → base)

Returns: index (`0x01` base, `0x03` FHR1, `0x08` FHR2, `0x0C` TOCO), bed number, 3-byte version, 4-byte build timestamp, 6-byte MAC.

Alarm Status (`0xD0`)

1 byte alarm level + 1 byte alarm volume.

Pairing Info (`0x08`)

- `0x16` = bed number, `0x17` = channel 1, `0x18` = channel 2.

3.6 Wired Probe Link (Serial Variant)

For wired probes connected directly to the receiving board:

Parameter	Value
Baud rate	19200
Frame	8 data bits, 1 stop bit, no parity
Heartbeat	`0xA1` every 1 s (bed number, probe type, battery)
Set parameters	`0xC2` (bed number, channel); probe acknowledges

4. Power and Battery Subsystem

4.1 Battery Chemistry and Cell

The probes and base use **lithium-ion polymer (LiPo)** rechargeable cells.

Parameter	Value
Supplier	Bohai Energy (Shenzhen) Co., Ltd.
Cell model	523450
Capacity	1000 mAh
Chemistry	$LiCoO_2$ cathode (~25–35%), graphite anode (~15–20%), PVDF binder (1–5%), acetylene black (0.5–3%), aluminum foil (~21–23%), copper foil (~10–11%), $LiPF_6$ electrolyte (~10–15%)
Nominal voltage	3.7 V (typical for 523450 format — confirm with supplier)
Shipping	UN 3480 / 3481; IATA PI 965–967; IMDG SP 188
Storage temperature	–20 °C to +35 °C (long-term ~20 ± 5 °C)
Critical limit	**Do not expose to >60 °C** — venting of flammable liquid/gas

> **Note.** The MSDS is supplied for regulatory and transport compliance. Do not reproduce it verbatim without supplier permission; reference it as *MSDS — Li-ion polymer cell 523450/1000mAh, Bohai Energy, effective 2024-01-01*.

4.2 Runtime and Charging

Component	Working time	Charge time
US probe	~11 h	~2 h
TOCO probe	~11 h	~2 h
Wireless base	~4.5 h	~4 h

4.3 Charging Behavior

- Docking a probe on the base rack **automatically turns the probe off** and starts charging.
- Charging indicator: lit during charge, off when complete.

- The base charges from the monitor's external AC supply while the monitor is on AC.
- **Do not** short battery terminals; do not reverse polarity; do not puncture, crush, or immerse.

4.4 Battery Safety Warnings

- Keep away from open flame and heat sources.
- Do not disassemble without authorization.
- Avoid dropping or mechanical impact.
- Dispose per local regulations; fully discharge and tape terminals before disposal.
- The cell contains recyclable materials — use a licensed waste carrier.

5. Environmental and Mechanical Characteristics

5.1 Environmental Limits

Parameter	Operating	Transport / Storage
Temperature	+5 °C to +40 °C	−10 °C to +55 °C
Humidity	< 85% RH	< 93% RH
Atmospheric pressure	86–106 kPa	86–106 kPa

5.2 Mechanical

Parameter	Value
Base dimensions	210 × 140 × 115 mm
Base weight	0.7 kg (incl. battery)
Base ingress	IPX1 (drip-proof)
Probe display	0.96" color OLED, 128×64
Case	Aluminum carry case

5.3 Electromagnetic and Interference Notes

- Keep the device away from strong EMI sources (wireless transmitters, electrosurgery, MRI).
- Do not use during electrosurgery or MRI — this is a **contraindication**.
- WLAN coexistence: use channels 84–99.
- Maintain 5 m max probe-to-base separation; avoid obstacles and hand-blocking of probes.

6. Operating Workflow

6.1 Pre-Monitoring Setup

1. Connect the wireless base to the monitor using the dedicated communication cable.
2. Power the monitor from external AC supply (required for base/probe charging and setup).
3. Power on the wireless base (press On/Off).
4. In the monitor's "Fetal and Pregnant Woman" menu, select **Wireless** as the probe type.

6.2 Base Configuration

6.2.1 Bed and Channel

1. Press **Setup** on the base.
2. Select **BED** or **channel** with Up/Down.
3. Press **Setup** to enter edit mode.
4. Set value with Up/Down.
5. Press **Back** to confirm and exit.

> **Important.** If the channel is changed, **re-set the corresponding probe** (§6.3).

6.2.2 Recommended Channel Map

Bed	Channel 1	Channel 2	Bed	Channel 1	Channel 2
1	—	—	9	—	—
2	93	97	10	84	88
3	87	91	11	92	96
4	—	—	12	—	—
5	—	—	13	—	—
6	85	89	14	94	98
7	95	99	15	86	90
8	—	—	16	—	—

6.3 Probe Pairing / Setup

US and TOCO probes

1. Place the probe to be set on its socket on the base rack.
2. Select the corresponding item with Up/Down: **US1**, **US2**, or **TOCO**.
3. Press **Setup** — all probe displays on the socket turn on.
4. When the display shows **SETUP OK**, the probe is paired.

FM marker

1. Select **FM** with Up/Down.
2. Press **Setup** and confirm.
3. When **"FM setting please wait"** appears, press the FM marker for **2 seconds**.
4. Restart the terminal to verify pairing.

6.4 Monitoring

1. Locate the position with the clearest fetal heart tones.
2. Fix the US probe with a strap; adjust tightness for patient comfort.
3. Fix the TOCO probe on the abdomen with a strap.
4. Press **Reset** on the monitor panel to establish the TOCO baseline.
5. Monitoring starts automatically once probes are fixed.
6. Hand the FM marker to the patient; she presses it on perceived fetal movement.

6.5 Shutdown

1. Dock the probes on the base rack (this turns them off).
2. Press and hold the base On/Off key for **3 seconds**.

7. Charging and Maintenance

7.1 Routine Charging

- Dock probes on the base rack; the base must be connected to the monitor with AC power applied.
- The base's own battery charges when the monitor is on AC.
- Charging indicators extinguish when complete.

7.2 Battery Replacement

> **Only qualified personnel** should replace batteries.

1. Power off the device and unplug AC.
2. Loosen the battery-cover screws and open the cover.
3. Unplug the battery connector and remove the cell.
4. Insert the replacement cell, reconnect the plug, and seat it in the compartment.
5. Replace the cover and tighten the screws.

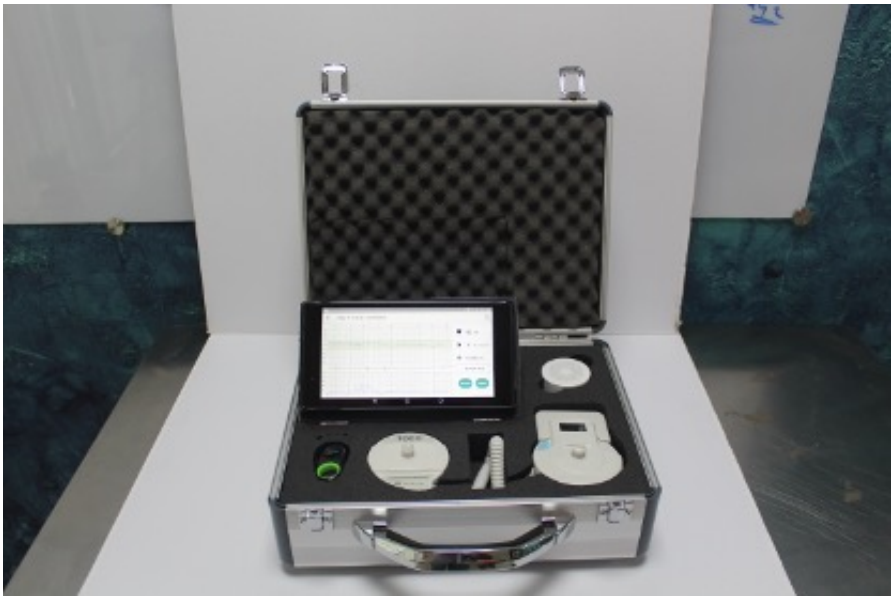
7.3 Preventive Maintenance

Task	Interval
Visual inspection for damage	Monthly or shorter
Functional safety check	Every 2 years or per local regulation
Firmware / software update	Per release
Battery health check	On each charge cycle

7.4 Cleaning

- Wipe exterior with a soft, damp cloth.
- **Do not** sterilize by high temperature, electron beam, or gamma radiation.
- **Do not** immerse in water (IPX1 only).
- **US probe:** water-based couplant only; wipe clean after use.
- **TOCO probe:** no couplant; keep liquids away.

8. Reference Bill of Materials



Complete Fetosense Kit in Aluminum Carry Case

#	Item	Qty	Key specification
1	Wireless US probe (FHR1)	1	1.0 / 2.0 MHz, 50–210 bpm, ±2 bpm
2	Wireless US probe (FHR2, twin option)	0–1	Same as FHR1
3	Wireless TOCO probe	1	0–100 units, ±10%
4	Wireless FM marker	1	Manual event input
5	Wireless base / receiving board	1	210×140×115 mm, 0.7 kg, IPX1
6	Communication cable	1	Base ↔ monitor (UART 115200 8N1)
7	Tablet (Lenovo M8)	1	Runs Fetosense Device app
8	Wireless Speaker	1	Audio output
9	Aluminum carry case	1	—
10	Li-ion polymer cell 523450	3–4	1000 mAh, one per probe + base
11	Water-based ultrasound couplant	1	US probe only
12	Straps / Abdominal Belt	2–3	For US and TOCO probes
13	Charging Adapter & Cables	2–3	USB power adapters and cables

9. Regulatory and Safety Context

9.1 Regulatory Classification (Reference Product Family)

Parameter	Value
Classification	Class IIa (Rule 10, Annex IX, MDD 93/42/EEC)
Conformity route	Annex II excluding (4)
Notified Body	MEDCERT GmbH (ID 0482)
EC Certificate	6016GB410190923
CE marking start	15 August 2018
Declaration of Conformity	27 September 2019
Models covered	L8D, L8D2, eFM-60, L8T31 (Reference family)

9.2 Applied Standards

- IEC 60601-1:2005+A1:2012 — General safety and essential performance
- EN IEC 60601-1-2:2015 — Electromagnetic compatibility
- IEC 60601-2-37:2007 — Ultrasonic diagnostic and monitoring equipment
- ISO 10993-1 / -5 / -10 — Biological evaluation
- EN 62479:2010 — Human exposure to EM fields (10 MHz–300 GHz)
- Harmonised standard for 2.4 GHz radio spectrum access
- MEDDEV 2.7/1 rev.4 — Clinical evaluation
- MEDDEV 2.12/1 rev.8 — Vigilance system

9.3 Intended Use and Indications

- **Intended use:** Non-invasive monitoring and recording of physiological parameters of the fetus in healthcare environments.
- **Indications:** NST from the 28th week of gestation; continuous and auscultatory monitoring of TOCO, FM, and FHR of single fetuses.
- **Users:** Trained and qualified personnel only.
- **Environments:** Antepartum examination rooms, gynecology clinics, pre-delivery rooms, labor and delivery rooms.

9.4 Contraindications

- **Not** for use during electrosurgery.

- **Not** for use during MRI.
- **Not** for intensive care units, operating rooms, or home use in the reference regulatory filing.

9.5 Essential Safety Warnings

1. Do not operate in flammable-gas environments.
2. One patient at a time.
3. Not defibrillation-proof.
4. Use only manufacturer-supplied probes.
5. Qualified maintenance only.
6. No high-temperature, e-beam, or gamma sterilization.
7. Avoid strong EMI sources.
8. Inspect for damage before each use; safety check every 2 years.
9. Water-based couplant only on US probe; none on TOCO.
0. Verify FHR with a stethoscope if the reading is not credible.
1. Keep probe within 5 m of the base.
2. Battery: no terminal shorting, no open flame, no unauthorized disassembly.

10. Building Your Own Device

10.1 What This Document Enables

A developer can use this document to:

- Build a **functionally equivalent research prototype** using the described architecture.
- Implement a **compatible host application** that speaks the protocol in §3.
- Design a **custom receiving board** that bridges BLE probes to a host.
- Integrate the reference kit with the Fetosense OSS software stack.

10.2 What This Document Does Not Grant

> **Critical:**
>
> - **No patent license.** Certain probe and signal-processing designs may be patented. Building a derivative for commercial use may require a separate license.
> - **No regulatory approval.** Any device intended for clinical use must undergo independent verification, validation, and regulatory clearance in the target jurisdiction.
> - **No clinical validation.** The clinical evidence summarized in §9 applies to the reference product family under its specific indications; it does not automatically extend to derivative devices.
> - **No third-party IP.** Protocol documents, safety data sheets, and supplier materials remain the property of their respective owners.

10.3 Recommended Build Approach

1. **Start with the protocol.** Implement §3 as a test harness against a reference base.
2. **Use off-the-shelf BLE modules** (e.g., a BLE-to-UART bridge) to emulate probe ↔ base links during development.
3. **Validate signal quality reporting** (Status bits 1–0) against known-good probes.
4. **Implement the ADPCM decoder** once the decode reference document is obtained (§11).
5. **Test battery reporting** across the full 0–100 range and at dock/undock transitions.
6. **Document deviations** from this reference architecture in your own repository.

11. Open Items and Requested Disclosures

The following items are **not fully covered** by currently available public documentation and should be requested from the OEM supplier or confirmed internally:

#	Item	Why it matters	Requested from
1	BLE GATT service and characteristic UUIDs	Direct probe integration without a base	OEM supplier
2	Probe electrical ratings (supply V, mA, charge voltage/current)	Custom base or charger design	OEM supplier
3	ADPCM decode reference (protocol LKN-WL0009)	Sound playback implementation	OEM supplier
4	Exact US frequency of the OSS-procured variant (1.0 vs 2.0 MHz)	Acoustic output and safety calculations	CareNX / OEM
5	Speaker wiring / audio source (tablet jack vs. base)	Kit assembly accuracy	CareNX
6	Redistribution permission for protocol and MSDS	Public repo compliance	OEM / supplier legal
7	Nominal cell voltage and charge profile	Battery safety analysis	Cell supplier
8	Whether the OSS app talks to the base (UART) or directly to probes (BLE)	Integration architecture	CareNX software team

12. Glossary

Term	Definition
ADPCM	Adaptive Differential Pulse-Code Modulation — audio compression used for sound packets
AFM	Automatic Fetal Movement
BLE	Bluetooth Low Energy
CTG	Cardiotocography — combined FHR and uterine contraction recording
FHR	Fetal Heart Rate
FM	Fetal Movement
GATT	Generic Attribute Profile — BLE data organization
ISM	Industrial, Scientific, Medical (2.4 GHz band)
NST	Non-Stress Test
OEM	Original Equipment Manufacturer
PCM	Pulse-Code Modulation
TOCO	Tocodynamometer — uterine contraction sensor
UART	Universal Asynchronous Receiver/Transmitter

13. References

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